Christopher Cain

160 Wileman Way, Cambridge, UK, CB3 1AR
clc218@cam.ac.uk • +44 07771 124954
ORCID: http://orcid.org/0000-0001-9420-7384
Website: https://clcain3.wixsite.com/christopher-cain
UAT Concepts: Reionization, Intergalactic Medium, Radiative Transfer

EMPLOYMENT

Senior Postdoctoral Fellow, Kavli Institute for Cosmology, Cambridge University Oct 2026 - Present

- I carry out an independent research program in collaboration with faculty, postdocs, and students at the Kavli Institute for Cosmology at Cambridge.

Beus Prize Postdoctoral Research Fellow, Arizona State University Sep 2023-Sep 2026

- Beus Center Director: Prof. Judd Bowman
- Independent postdoctoral researcher. I carry out research projects in collaboration with faculty, postdocs, and graduate students in and outside the department, mentor undergraduate students, and present at conferences and workshops.

Graduate Research Assistant, University of California, Riverside Jan 2020-Jul 2023

- Supervisor: Prof. Anson D'Aloisio (Ph.D. advisor)
- I carried out research projects as part of my Ph.D. work, helped mentor other graduate students and undergraduates, TA'ed for labs, and presented my work at conferences.

Undergraduate Research Assistant, University of Oklahoma May 2017-Aug 2018

- Supervisor: Prof. Eddie Baron
- I was a Research Experience for Undergraduates (REU) student during the summer of 2017 ($5500 award), and continued my work through the following year after the end of the REU.

EDUCATION

University of California, Riverside, Riverside, California, USA

- Ph.D. in Physics, specialization in Astrophysics Jan 2020-Jul 2023
  - Thesis Title: *Modeling Reionization from Small to Large Scales*
- M.S. in Physics Sep 2018 - Jan 2020

Azusa Pacific University, Azusa, California, USA

- B.S. in Physics, B.S. in Mathematics (Summa Cum Laude) Sep 2014 - Dec 2017

RESEARCH INTERESTS

- Computational modeling of the Epoch of Reionization using radiative transfer simulations and the development of efficient theoretical tools to interpret observations probing the early universe.
- Small-scale structure and dynamics of the intergalactic medium.
- The evolution of the first galaxies, black holes, and stars in the first billion years.
- Synergies between high-redshift astrophysics and precision cosmology.

SCHOLARSHIPS FELLOWSHIPS, & AWARDS

- 1st Place 2025 Buchalter Cosmology Prize Jan 2026
  - Annual prize that seeks to stimulate ground-breaking theoretical, observational, or experimental work in cosmology that has the potential to produce a breakthrough advance in our understanding. It was created to support the development of new theories, observations, or methods, that can help illuminate the puzzle of cosmic expansion from first principles.
- Beus Center - Director's Award for Excellence in Mentoring Aug 2025
  - Beus Center for Cosmic Foundations, School of Earth & Space Exploration, ASU
- Robert T. Poe Memorial Scholarship for Outstanding PhD Graduate Jun 2023
  - Dept. of Physics and Astronomy, University of California Riverside
- Beus Prize Postdoctoral Fellowship, Feb 2023
  - Beus Center for Cosmic Foundations, School of Earth & Space Exploration, ASU
- Senior Graduate Student of the Year Award, Jun 2022

- Dept. of Physics and Astronomy, University of California Riverside
- Junior Graduate Student of the Year Award, Jun 2020
  - Dept. of Physics and Astronomy, University of California Riverside
- 1st Year Graduate Student of the Year Award Jun 2019
  - Dept. of Physics and Astronomy, University of California Riverside
- Provost Research Fellowship, University of California Riverside Apr 2018

AWARDED COMPUTE ALLOCATIONS

- NSF ACCESS Accelerate Compute Allocation (PHY250179, Co-I [PI Matthew McQuinn])
  High-resolution cosmological simulations of intergalactic gas
  - $\approx$ 3 million CPU hours + 50 TB storage, est. value (to-date) $5,630.00 Jun 2025-Present
- NSF ACCESS Explore Compute Allocation (PHY250106, Co-I [PI Kevin Croker])
  Impacts of Cosmologically Coupled Black Holes on Supernovae and Large-Scale Structure Observables
  Apr 2024-Present
- NSF ACCESS Accelerate Compute Allocation (PHY240332, PI)
  An accurate parameter study of reionization to constrain the ionizing properties of the first galaxies
  - $\approx$ 2.5 million CPU hours + 450 TB storage, est. value (to date) $22,760.00 Jan 2025-Present
- NSF ACCESS Accelerate Compute Allocation (PHY230158, PI)
  A comprehensive treatment of the intergalactic ionizing opacity for studies of reionization
  - $\approx$ 3 million CPU hours + 400 TB storage, est. value $92,720 Oct 2023-Dec 2024
- NSF ACCESS Explore Compute Allocation (PHY210041, Co-I [PI Anson D'Aloisio])
  Simulating Reionization in the Era of JWST and Extremely Large Telescope
  - $\approx$ 400K CPU hours per year Jul 2021-Present
- NSF ACCESS Explore Compute Allocation (PHY230063, PI)
  Accurate simulations of reionization for studies of the Lyman alpha forest and other high-redshift observables May 2023-May 2024

AWARDED PROPOSALS

- George Becker, ...**Christopher Cain**, et. al., "The Mean Free Path at z = 5.6: Insights into How Reionization Ends", ESI proposal 2024A U281 Sep 2023
- Yongda Zhu, ...**Christopher Cain**, et. al., "The Mean Free Path of Ionizing Photons at z = 5.6: A Robust Constraint on Reionization", ALMA proposal 2022.1.00662.S Apr 2022
- George Becker, ...**Christopher Cain**, et. al., "The Mean Free Path at z = 5.6: Insights into Ultra-Late Reionization", Keck proposal 2021B_U036 Mar 2021

UNDERGRADUATE STUDENTS ADVISED

- **Joshua Cohon** Oct 2024 – Apr 2026
  - ASU undergrad researcher - he studied the implications of a recent detection of a Lyα emitting galaxy by JWST at $z = 13$. He presented his work at the 246th AAS meeting and wrote a lead-author paper. He is now working on IGM magnetic fields, and was recently awarded a prestigious Goldwater scholarship for his proposed work.
- **Alexandra Nelander** Jun 2024 – Jul 2025
  - ASU undergraduate researcher working on her senior thesis. Her project is about the relationship between high-redshift AGN and the EDGES cosmological 21 cm detection. She wrote and defended her undergraduate thesis on this work, and produced a lead-author publication.
- **Esteban Moreno** Oct 2023 – Present
  - ASU undergraduate researcher. He is studying the Lyα and Lyman series transmission properties of the IGM. He recently completed his Senior Thesis.
- **Aloha Das** Sep 2023 – Present
  - ASU undergrad researcher studying the relationship between ionizing photon absorbers and dark matter halos. He is currently almost finished with a first-author paper.
- **Joshua Roth** Jun 2022 – Jul 2023
  - UCR Undergraduate researcher. He presented results at the 241st AAS meeting, published a lead-author paper in June, 2024, and was accepted to Princeton for grad school.
- **Andrew Caruso** Jan 2023 – Jun 2023
  - UCR undergraduate who completed his senior thesis at the end of Spring 2024 under my supervision.
- **Sanya Dhami** Jun 2021 – Mar 2022
  - UCR Undergraduate researcher during the summer and academic year. She contributed significantly to and was co-author on my paper "Small-scale clumping of dark matter and the mean free path of ionizing photons at $z = 6$".

GRADUATE STUDENTS MENTORED

- **Joseph Thornton** Mar 2026 – Present
  - Ph.D. Student at Cambridge University working under Blake Sherwin. His work focuses on using the kSZ effect from the CMB to constrain reionization.
- **Yash Aggarwal** Sep 2025 – Present

- PhD Student at UC Riverside Prof. Anson D'Aloisio. His research includes a study of the effect of Pop III.1 stars on high-redshift observables that probe reionization.

- **Hurum Tohfa** Sep 2023 – Feb 2026
  - PhD student at the University of Washington advised by Prof. Matthew McQuinn. She recently concluded a study of the mean free path of ionizing photons during reionization.
- **Garett Lopez** Sep 2023 – Jun 2026
  - PhD student at UC Riverside advised by Prof. Anson D'Aloisio. His research focuses on modeling CMB observables that probe reionization using RT simulations.

PUBLICATIONS

Total *published* papers: 35 (+6 submitted, +6 in prep., 47 total)
Single/First-author papers: 15 (+1 submitted, +1 in prep., 17 total)
First-authored by advised/mentored students: 5 (+1 submitted, +1 in prep., 7 total)
Other 2nd/Nth author papers: 15 (+4 submitted, +4 in prep., 23 total)
Total (first-author) citations: 762 (297) Source: NASA ADS
H-index: 15 M-index: 1.7 G-index: 27 I10-index: 19 RIQ-index: 157 Tori-index: 2.0
First-Author Only H-Index: 9

NASA ADS Link

Single-Author

- **Christopher Cain**, "Towards an accurate treatment of the reduced speed of light approximation in parameterized radiative transfer simulations of reionization", *Journal of Cosmology and Astroparticle Physics*, vol. 2024, no. 12, Dec. 2024 [**cited: 6]**

First-Author

- **Christopher Cain**, Kimberly K. Boddy, Simeon Bird, Anson D'Aloisio, Hy Trac, "(Non-)Convergence of the Small-Scale Lyα Forest Flux Power Spectrum at $z = 4{-}5$ in Blue-Tilted Cosmologies", draft in co-author review.
- **Christopher Cain**, Brent Smith, Gibson B. Bowling, Yongda Zhu, Lily Whitler, Rafael Ortiz III, Anson D'Aloisio, Rogier Windhorst, "Super-Eddington Little Blue Dots May Reionize Helium Too Early", submitted to ApJ Letters.
- **Christopher Cain**, Kevin S. Croker, Anson D'Aloisio, Ivelin Georgiev, Hurum M. Tohfa, Yongda Zhu, and Rogier Windhorst, "A self-consistent analytical model for both the ionizing background and reionization history", accepted to JCAP. [**cited: 1]**
- **Christopher Cain**, Aloha Das, Anson D'Aloisio, Matthew McQuinn, Evan Scannapieco et. al., "Introducing SAGUARO - simulating IGM evolution and environments at high resolution: setup and first results", *Journal of Cosmology and Astroparticle Physics*, vol. 2026, no. 19, pp. 24,Sep. 2026 [**cited: 5]**
- [*] **Christopher Cain**, Alexander Van Engelen, Kevin S. Croker, Darby Kramer, Anson D'Aloisio, and Garett Lopez, "The CMB optical depth constrains the duration of reionization", *The Astrophysical Journal Letters*, vol. 987, no. 2, pp. L29, Jul 2025. [**cited: 30]**
- [*] **Christopher Cain**, Matthew McQuinn, Evan Scannapieco, Anson D'Aloisio, and Hy Trac, "Kiloparsec-scale turbulence driven by reionization may grow intergalactic magnetic fields", *Physical Review D*, vol. 113, no. 2, pp. 023050, Jan. 2026 [**cited: 4]**
- **Christopher Cain**, Anson D'Aloisio, and Julian Munoz, "New constraints on the galactic ionising efficiency and escape fraction at $2.5 < z < 6$ based on quasar absorption spectra", *Publications of the Astronomical Society of Australia*, vol. 42, pp.107, Jul 2025 [**cited: 13]**
- **Christopher Cain**, Garett Lopez, Anson D'Aloisio, Julian Munoz, Rolf Jansen, Rogier Windhorst, and Nakul Gangolli, "Chasing the beginning of reionization in the JWST era", *The Astrophysical Journal*, vol. 980, no. 1, pp. 83, Feb 2025. **[cited: 32]**
- [*] **Christopher Cain** & Anson D'Aloisio, "FlexRT - A fast and flexible cosmological radiative transfer code for reionization studies I: Code validation", *Journal of Cosmology and Astroparticle Physics*, vol. 2024, no. 12 Dec 2024. **[cited: 16]**

[*] **Highest-importance publication.**

- **Christopher Cain**, Evan Scannapieco, Matthew McQuinn, Anson D'Aloisio, and Hy Trac, "The hydrodynamic response of small-scale structure to reionization drives large IGM temperature fluctuations that persist to z = 4", *Monthly Notices of the Royal Astronomical Society Letters*, vol. 533, no. 1, pp. L100, Jul 2024. **[cited: 11]**
- **Christopher Cain**, Anson D'Aloisio, Garett Lopez, Nakul Gangolli, and Joshua T. Roth, "On the rise and fall of galactic ionizing output at the end of reionization", *Monthly Notices of the Royal Astronomical Society*, vol. 531, no. 1, pp. 1951, May 2024. **[cited: 23]**
- **Christopher Cain**, Anson D'Aloisio, Nakul Gangolli and Matthew McQuinn, "The Morphology of Reionization in a Dynamically Clumpy Universe", *Monthly Notices of the Royal Astronomical Society*, vol. 522, no. 2, pp. 2047, Jun 2023 **[cited: 32]**
- **Christopher Cain**, Anson D'Aloisio, Vid Irsic, Nakul Gangolli, and Sanya Dhami, "Small-scale clumping of dark matter and the mean free path of ionizing photons at $z = 6$", *Journal of Cosmology and Astroparticle Physics*, vol. 2023, no. 1, Jan. 2023 **[cited: 9]**
- [*] **Christopher Cain**, Anson D'Aloisio, Nakul Gangolli, and George D. Becker, "A Short Mean Free Path at z = 6 Favors Late and Rapid Reionization by Faint Galaxies", *The Astrophysical Journal Letters*, vol. 917, no. 2, pp. 37, Aug 2021. **[cited: 85]**
- **Christopher Cain**, Anson D'Aloisio, Vid Irsic, Matthew McQuinn, and Hy Trac, "A Model-Insensitive Baryon Acoustic Oscillation Feature in the 21 cm Signal from reionization", *The Astrophysical Journal*, vol. 898, no. 2, pp. 168, Aug 2020. **[cited: 23]**
- [†] **Christopher Cain**, E. Baron, M. M. Phillips, C. Contreras, C. Ashall, et al., "Investigating the Unusual Spectroscopic Time-Evolution in SN 2012fr," *The Astrophysical Journal*, vol. 869, no. 2, pp. 162, Dec 2018. **[cited: 7]**

First-Authored by Advised Undergraduates

- Joshua Cohon, **Christopher Cain**, Rogier Windhorst, Anson D'Aloisio, Timothy Carleton, and Yongda Zhu, "A long time ago in an LAE far, far away: a signpost of early reionization or a nascent AGN at $z = 13$?", *Publications of the Astronomical Society of Australia*, vol. 43, pp. e046, Mar. 2026 **[cited: 11]**
- Alexandra Nelander, **Christopher Cain**, Jordan C. J. D'Silva, Peter H. Sims, Rogier Windhorst, and Judd D. Bowman, "Can high-redshift AGN observed by JWST explain the EDGES absorption signal?", *Publications of the Astronomical Society of Australia*, vol. 43, pp. e037, Apr. 2026 **[cited: 4]**
- Joshua Roth, Anson D'Aloisio, **Christopher Cain**, Bayu Wilson, Yongda Zhu, and George D. Becker, "The effect of reionization on direct measurements of the mean free path", *Monthly Notices of the Royal Astronomical Society*, vol. 530, no. 4, pp. 5209, Jun 2024 **[cited: 21]**

First-Authored by Mentored Graduate Students

- Joseph Thornton, **Christopher Cain**, Niall MacCrann, Alexander Van Engelen, and Blake Sherwin, "A persistent τension: Constraints on the CMB optical depth with kSZ 2pt+4pt limits and the Lyα forest", in co-author review.
- Yash Aggarwal, **Christopher Cain**, Garett Lopez, Hy Trac, Anson D'Aloisio, Philip Tanedo, and Jonathan C. Tan, "Fireworks at Cosmic Dawn: resolving BAO-CMB tensions with the Pop III.1 flash", submitted to ApJ. [**cited: 4]**
- Hurum Tohfa, **Christopher Cain**, Matthew McQuinn, and Anson D'Aloisio, "An emulator for the ionizing photon mean free path in ultra-high resolution simulations: the implications of mean free path measurements for the reionization history", *Journal of Cosmology and Astroparticle Physics*, vol. 2026, no. 6, pp. 094, June 2026 **[cited: 2]**
- Garett Lopez, Anson D'Aloisio, and **Christopher Cain**, "Predicting the patchy kSZ signal in light of recent QSO absorption results", *Journal of Cosmology and Astroparticle Physics*, vol. 2026, no. 5, pp. 074, May 2026 **[cited: 2]**

[†]Completed as an undergraduate.

Other 2nd Author

- Niall MacCrann, **Christopher Cain**, Aleksandra Kusiak, Fiona McCarthy,, Alexander Van Engelen, Darby Kramer, William R. Coulton, Frank J. Qu, "Characterising the epoch of reionisation using the cross-correlation of the kSZ effect and CMB lensing", submitted to MNRAS.
- Fahad Nasir, **Christopher Cain**, Anson D'Aloisio, Nakul Gangolli, and Matthew McQuinn, "Hydrodynamic Response of the Intergalactic Medium to Reionization II: Physical Characteristics and Dynamics of Ionizing Photon Sinks", *The Astrophysical Journal*, vol. 923, no. 2, pp. 161, Dec 2021. **[cited: 30]**

Nth-Author with Considerable Contribution

- Yongda Zhu, Eiichi Egami, Marcia J. Rieke, Xiaohui Fan, George D. Becker, **Christopher Cain**, et. al., "Evidence for Late Reionization from Broadband Lyman-Series Opacity in Deep JWST/NIRCam Imaging", in co-author review.
- Garett Lopez, Anson D'Aloisio, and **Christopher Cain**, "Constraints on Reionization from Planck Low-ℓ EE Polarization and the Lyman-Series Forests", in co-author review.
- Yongda Zhu, Xiaohui Fan, Laura C. Keating, George D. Becker, Eiichi Egami,...,**Christopher Cain**, et. al., "Low Ly α Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over 4.8<z<11", submitted to ApJ.
- Darby Kramer, Alexander Van Engelen, **Christopher Cain**, Niall MacCrann, Hy Trac, Skylar Grayson, Evan Scannapieco, and Blake Sherwin, "Cross-correlating the patchy screening and kinetic Sunyaev-Zel'dovich effects as a new probe of reionization" *The Astrophysical Journal*, vol. 994, no. 1, pp. 82, Oct 2025. **[cited: 8]**
- Nakul Gangolli, Anson D'Aloisio, **Christopher Cain**, George D. Becker, and Holly Christenson, "The correlation between galaxy density and Lyα forest transmission in late reionization models ", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 3, pp. 69, Mar. 2025 **[cited: 15]**
- Bayu Wilson, Anson D'Aloisio, George D. Becker, **Christopher Cain**, and Eli Visbal, "Imaging reionization's last phases with I-front Lyman-α emissions", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 1, pp. 66, Jan. 2025 **[cited: 3]**
- Bayu Wilson, Anson D'Aloisio, **Christopher Cain**, Eli Visbal, and George D. Becker, "Quantifying Lyman-α emissions from reionization fronts", *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 1, pp. 65, Jan. 2025 **[cited: 4]**
- Piyanat Kittiwisit, Steven G. Murray, Hugh Garsden, Philip Bull, **Christopher Cain**, et. al., "matvis: A matrix-based visibility simulator for fast forward modelling of many-element 21 cm arrays", *RAS Techniques and Instruments*, vol. 4, pp. rzaf001, Jan. 2025 **[cited: 15]**
- Yongda Zhu, George Becker, Sarah Bosman, **Christopher Cain**, Laura Keating, et. al., "Damping Wing-Like Features in the Stacked Lyα Forest: Potential Neutral Hydrogen Islands at $z < 6$", *Monthly Notices of the Royal Astronomical Society Letters*, vol. 533, no. 1, pp. L49, Sept 2024 **[cited: 80]**
- Anson D'Aloisio, Matthew McQuinn, Hy Trac, **Christopher Cain**, and Andrei Mesinger, "Hydrodynamic Response of the Intergalactic Medium to Reionization", *The Astrophysical Journal*, vol. 898, no. 2, pp. 149, Aug 2020. **[cited: 83]**

Other Nth-Author

- Gibson B. Bowling, Charles L. Steinhardt, Lachlan J. Hobart, Carter Meyerhoff, Jackson Headon, **Christopher Cain**, et. al., "Physically Motivated Stellar Populations for Galaxies at Cosmic Dawn", in co-author review.
- Anson D'Aloisio, Nayan A. Camalon, L. Y. Aaron Yung, Rachel S. Somerville, Christopher Cain, and Nakul Gangolli, "Models of the declining ionizing emissivity near the end of reionization", in co-author review.
- Kevin S. Croker, Joshua Cohon, Viktor T. Toth, Massimiliano Rinaldi, Nicolas Fernandez,...,**Christopher Cain**, et. al., "A Strong Cosmologically Coupled Mass Growth Solution in Linearized General Relativity", submitted to PRD Letters.

- Darby M. Cramer, Alexander Van Engelen, Frank J. Qu, **Christopher Cain**, Irene Abril-Cabezas, et. al., "The Atacama Cosmology Telescope: Constraints on the anisotropic screening and birefringence effects with DR6", submitted to ApJ. [**cited: 1**]
- Yongda Zhu, George D. Becker, Anson D'Aloisio, Ryan Endsley, Nakul Gangolli, **Christopher Cain**, et. al. "Galaxy Underdensities Host the Clearest IGM Lyα Transmission and Indicate Anisotropic Reionization", *The Astrophysical Journal*, vol. 1002, no. 1, pp. 93, May 2026 **[cited: 4]**
- Yongda Zhu, Eiichi Egami, Xiaohui Fan, Fengwu Sun, George D. Becker, **Christopher Cain**, et. al., "Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$", *The Astrophysical Journal Letters*, vol. 995, no. 1, pp. L5, Dec 2025. **[cited: 7]**
- Yongda Zhu, Marcia J. Rieke, Luis C. Ho, Yang Sun, George H. Rieke, ..., **Christopher Cain**, et. al., "A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars", *The Astrophysical Journal*, vol. 1000, no. 2, pp. 312, Apr. 2026 **[cited: 6]**
- Yongda Zhu, Tom J. L. C. Bakx, Ryota Ikeda, Hideki Umehata, George D. Becker, **Christopher Cain**, et. al., "Discovery of a Unique Close Quasar–DSFG Pair Linked by a [C II] Bridge at $z = 5.63$", *Research Notes of the American Astronomical Society*, vol. 8, no. 11, pp. 284, Nov. 2024 **[cited: 6]**
- Geoff G. Murphy, Philip Bull, Mario G. Santos, Zara Abdurashidova, Tyrone Adams, ..., **Christopher Cain**, et. al., "Bayesian estimation of cross-coupling and reflection systematics in 21cm array visibility data", *Monthly Notices of the Royal Astronomical Society*, vol. 534, no. 3, pp. 2653, Nov. 2024 **[cited: 10]**
- Yongda Zhu, George D. Becker, Holly M. Christenson, Anson D'Aloisio, Sarah E. I. Bosman, ..., **Christopher Cain**, et. al., "Probing Ultra-late Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$", *The Astrophysical Journal*, vol. 955, no. 2, pp. 161, Oct 2023. **[cited: 91]**
- † Carlos Contreras, M. M. Phillips, Christopher R. Burns, Anthony L. Piro, B. J. Shappee, ..., **Christopher Cain** et. al., "SN 2012fr: Ultraviolet, Optical, and Near-Infrared Light Curves of a Type Ia Supernova Observed Within a Day of Explosion," *The Astrophysical Journal*, vol. 859, no. 1, pp. 1–24, May 2018. **[cited: 58]**

INVITED TALKS/SEMINARS (11 TOTAL)

- "Revealing the High-Redshift Universe with Cosmic Reionization", invited seminar at the Beus Center for Cosmic Foundations, Arizona State University Sep 2026
- "Tau and Turbulence: The importance of reionization at the largest and smallest scales", VICO-CICO-CASSUM Spring 2026 Workshop May 2026
- "Tau and Turbulence: Understanding reionization at the largest and smallest scales", KIPAC Tea Talk at Stanford Oct 2025
- "The effect of reionization on small-scale structure in the intergalactic medium", invited talk at IGM-Galaxy workshop in Trieste, Italy May 2025
- "Revealing the physics of reionization with high-redshift quasars, JWST observations, & simulations of the IGM", DUSC Seminar at the University of Washington Apr 2025
- "Revealing the timeline and sources of reionization with high-redshift quasars and JWST observations", seminar at the University of Arizona Mar 2025
- "Modeling Reionization from Small to Large Scales", DAWN Center, Copenhagen, Denmark Dec 2023
- "Modeling Reionization from Small to Large Scales", Seminar at UCLA Apr 2023
- "Modeling Reionization from Small to Large Scales", Seminar at UC Santa Cruz Mar 2023
- "Modeling Reionization from Small to Large Scales", Seminar at UC Santa Barbara Mar 2023
- † "Analysis of Unusual Si II Features of SN2012fr Using SYNOW and W7," *Carnegie Supernova Project Fall Meeting*, Pasadena, California, USA Oct 2017.

CONTRIBUTED TALKS (18 TOTAL)

- "Do Little Blue Dots make too many Little Blue Photons?", contributed talk at SESE Internal Symposium, Arizona State University Aug 2026
- "Tau and Turbulence: Understanding reionization at the largest and smallest scales", Cosmology in the Alps 2026, Les Diablerets, Switzerland Mar 2026
- "Tau and Turbulence: Understanding reionization at the largest and smallest scales", Arizona Lensing Day Workshop Sep 2025

- "Tau and Turbulence: Understanding reionization at the largest and smallest scales", Kaba Kada Conference, Cairns, Australia Sep 2025
- "The CMB Optical Depth Constrains the Duration of Reionization", contributed talk at SESE Internal Symposium, Arizona State University Aug 2025
- "New constraints on the galactic ionizing efficiency and escape fraction at $2.5 < z < 6$ based on quasar absorption spectra", talk at AAS 246th AAS meeting, Anchorage, Alaska Jun 2025
- "Measuring the average escape fraction with high-z quasar data + HST/JWST", Cosmic Frontier Center conference at UT Austin May 2025
- "Chasing the beginning of reionization in the JWST era", 40th IAP Symposium, Paris, France Dec 2024
- "Chasing the beginning of reionization in the JWST era", Arizona Lensing Day Workshop Dec 2024
- "A tale of three histories: distinguishing reionization scenarios in the JWST Era", lightning talk at Kalvi Institute, UC Santa Barbara. Aug 2024
- "A tale of three histories: distinguishing reionization scenarios in the JWST Era", talk at *Cosmic Dawn at High Latitudes Conference*, Stockholm, Sweden Jun 2024
- "Studying the high-redshift galaxy population with reionization", Arizona Lensing Day Workshop Dec 2024
- "Ending Reionization Gracefully: Properties of the ionizing photon sources at the end of reionization", talk at *Reionisation in the Summer* Conference, Heidelberg, Germany Jun 2023
- "Small-scale clumping of dark matter and the mean free path of ionizing photons at $z = 6$", talk at UCLA Dark Matter 2023 conference Mar 2023
- Lightning talk at *Reionization on a Blackboard Workshop* at Flatiron Institute, NYC Sep 2022
- 'The optically thick IGM at $z \geq 6$: Improved modeling techniques and implications for observations", talk at *Reionization and Cosmic Dawn: Looking Forward to the Past* at UC Berkeley, CA Mar 2022
- "A short mean free path at z = 6: implications for reionization", Live online talk at *SAZERAC Reionization Conference* Jun 2021
- "A Model-Insensitive Baryon Acoustic Oscillation Feature in the 21 cm Signal from reionization," Recorded talk at *SAZERAC Reionization Conference* Jun 2020

SKILLS

[‡]**C/C++**, **Fortran**, **Python**, **Bash Script**, MATLAB, Mathematica, R, Java, **LaTeX**, Microsoft Office, LibreOffice, **Linux**, Windows

SOFTWARE EXPERIENCE

- [§]*FlexRT*: 3D Cosmological Radiative Transfer Code for Reionization Studies (C++)
- [§]1D Radiative Transfer & Hydrodynamics Code (C++)
- [§]Code for calculating Lyman Series transmission spectra in cosmological simulations (C++)
- [¶] RadHydro Reionization RT Code (Fortran, Author: Hy Trac)
- [¶]P^{3}M Cosmological N-body Code (Fortran, Author: Hy Trac)
- [¶]*HealPIX*, Hierarchical Equal Area isoLatitude Pixelization, (Python/C++/Fortran, various authors)
- [¶]*POWERS*, code for calculating the angle-averaged power spectrum of a field (Fortran, Author: Yi Mao)
- *MP-Gadget*, Cosmological N-body/Hydrodynamics code (C++, Authors: Yu Feng, Simeon Bird)
- *AMBER*, Semi-numerical reionization code (Fortran, Author: Hy Trac)
- *MATVIS*, interferometric visibility simulator (Python, Authors: HERA team)

COMMUNITY & PROFESSIONAL SERVICE

Academic Journals

- Referee for *The Astrophysical Journal*, *Monthly Notices of the Royal Astronomical Society*, *Journal of Astroparticle Physics & Cosmology*, *Physical Review D*, *The Open Journal of Astrophysics* (14 papers)

Proposal Review

- Proposal reviewer for the NASA Postdoctoral Program. 2025/2026
- Reviewer for computer time proposals for the DiRAC Resource Allocation Committee. 2025

Committees

- ASU School of Earth & Space Exploration Internal Research Symposium Jun 2024 – Aug 2024
- Beus Center for Cosmic Foundations Seminar Series Sep 2023 – Present

[‡]Boldface indicates extensive experience.
[§]Indicates fully original software.
[¶]Indicates extensive experience and/or made significant modifications.

Outreach

- Contributor to *Astronomy Magazine* (https://www.astronomy.com)
- Student Volunteer, *Dept. of Mathematics and Physics*, Azusa Pacific University Jan 2015 – Dec 2017
  - Assisted various department activities and community outreach programs.

TEACHING EXPERIENCE

University of California, Riverside, Riverside, California, USA

- Teacher's Assistant, Department of Physics and Astronomy Sep 2019 – Jun 2020
  - Courses Taught: PHYS 2LA, 2LB, 2LC (undergraduate physics labs)

Azusa Pacific University, Azusa, California, USA

- Tutor, Dept. of Mathematics and Physics Sep 2015 – Dec 2017
- Teacher's Assistant, Dept. of Mathematics and Physics Jan 2016 – Dec 2017
  - Courses taught: PHYS 151 (Introductory Physics), MATH 280 (Discrete Math & Proof)

Other

- Private Math & Physics Tutor
  - Over 1200 hours of private tutoring experience logged independently and on platforms such as Wyzant and Chegg.

PROFESSIONAL REFERENCES

Prof. Anson D'Aloisio (Ph.D. Advisor)
Associate Prof. of Physics & Astronomy
University of California, Riverside
900 University Ave, Riverside, CA 92521
ansond@ucr.edu

Prof. Matthew McQuinn (Letter Writer)
Associate Prof. of Physics & Astronomy
University of Washington
University 1900 Commerce St., Seattle, WA 98195
mcquinn@uw.edu

Prof. Judd Bowman (Beus Center Director)
Professor, School of Earth & Space Exploration
Arizona State University
1151 S Forest Ave, Tempe, AZ 85287
Judd.Bowman@asu.edu

Prof. Alexander Van Engelen (Letter Writer)
Assistant Prof., School of Earth & Space Exploration
Arizona State University
1151 S Forest Ave, Tempe, AZ 85287
alexander.van.engelen@asu.edu

Prof. Rogier Windhorst (Letter Writer)
Regents' Professor
Arizona State University
1151 S Forest Ave, Tempe, AZ 85287
Marisol.J.Diaz@asu.edu

Prof. Bradley McCoy (Reference)
Professor, SCEMPS Department
Azusa Pacific University
901 E Alosta Ave, Azusa, CA 91702
bmccoy@apu.edu